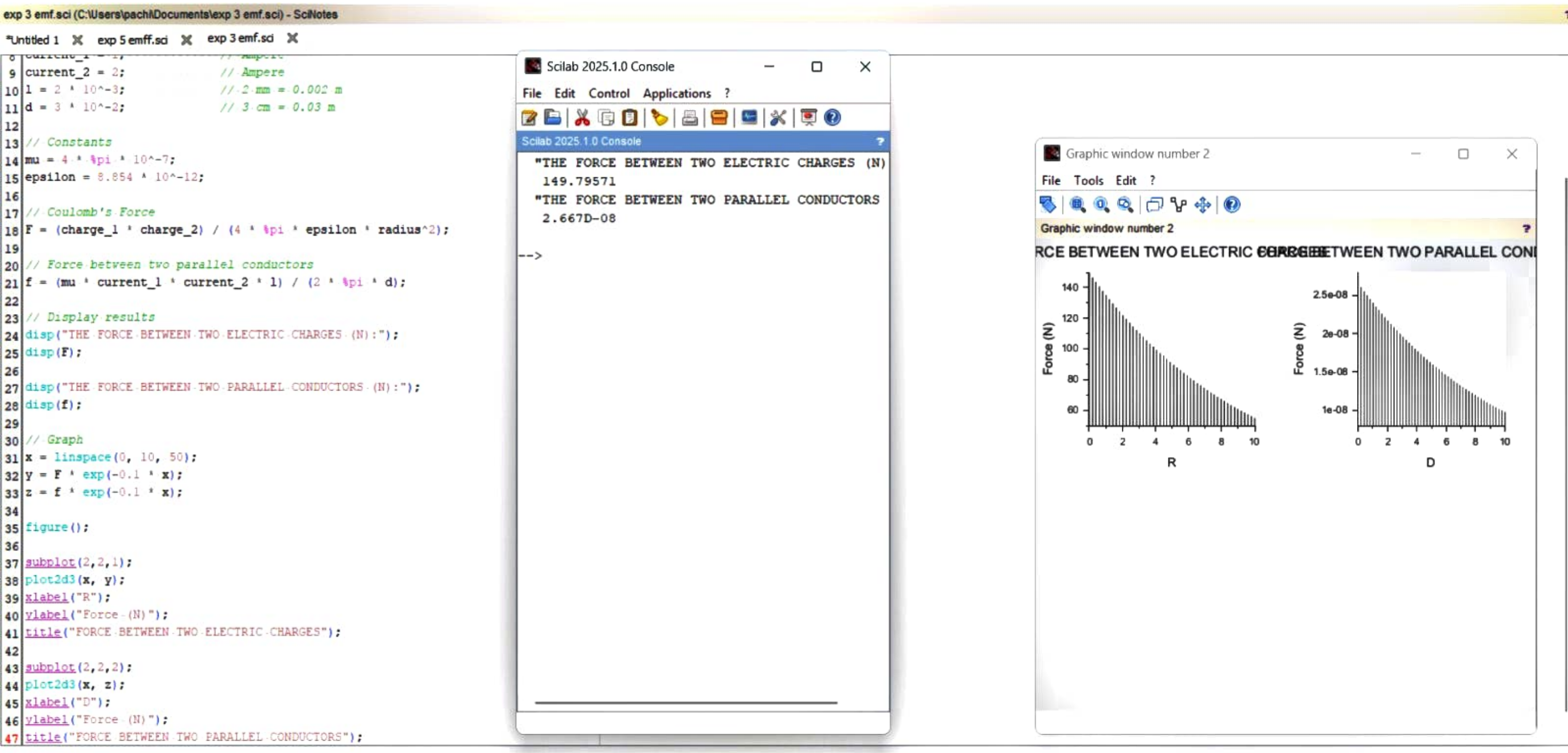
exp 3 emf.sci (C:\Users\pachi\Documents\exp 3 emf.sci) - SciNotes
*Untitled 1
exp 5 emff.sci
exp 3 emf.sci
current_2 = 2; // Ampere
l = 2 * 10^-3; // 2 mm = 0.002 m
d = 3 * 10^-2; // 3 cm = 0.03 m
// Constants
mu = 4 * %pi * 10^-7;
epsilon = 8.854 * 10^-12;
// Coulomb's Force
F = (charge_1 * charge_2) / (4 * %pi * epsilon * radius^2);
// Force between two parallel conductors
f = (mu * current_1 * current_2 * l) / (2 * %pi * d);
// Display results
disp("THE FORCE BETWEEN TWO ELECTRIC CHARGES (N):");
disp(F);
disp("THE FORCE BETWEEN TWO PARALLEL CONDUCTORS (N):");
disp(f);
// Graph
x = linspace(0, 10, 50);
y = F * exp(-0.1 * x);
z = f * exp(-0.1 * x);
figure();
subplot(2,2,1);
plot2d3(x, y);
xlabel("R");
ylabel("Force (N)");
title("FORCE BETWEEN TWO ELECTRIC CHARGES");
subplot(2,2,2);
plot2d3(x, z);
xlabel("D");
ylabel("Force (N)");
title("FORCE BETWEEN TWO PARALLEL CONDUCTORS");
Scilab 2025.1.0 Console
File Edit Control Applications ?
"THE FORCE BETWEEN TWO ELECTRIC CHARGES (N)
149.79571
"THE FORCE BETWEEN TWO PARALLEL CONDUCTORS
2.667D-08
-->
Graphic window number 2
File Tools Edit ?
RCE BETWEEN TWO ELECTRIC CHARGES
FORCE BETWEEN TWO PARALLEL CONI
Force (N)
R
D

T[illegible] Exp 3:-

$q_1 = 5 \times 10^{-6}$

$q_2 = 3 \times 10^{-6}$

$r = 3\ cm$

$I_1 = 1\ \cancel{amp}$

$I_2 = 2\ \cancel{amp}$

$l = 2\ mm$

$d = 3\ cm$

* Force b/w Two electric

$$F = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2}$$

$$q_1 q_2 = (5 \times 10^{-6})(3 \times 10^{-6}) = 15 \times 10^{-12} = 1.5 \times 10^{-11}$$

$$r^2 = (0.03)^2 = 0.009$$

$$4\pi\varepsilon_0 r^2 = (4 \times 3.1416)(8.854 \times 10^{-12})(0.009)$$

$$= 12.5664 \times 8.854 \times 10^{-12} \times 0.009$$

$$= 1.0016 \times 10^{-13}$$

$$F = \frac{1.5 \times 10^{-11}}{1.0016 \times 10^{-13}}$$

$$\boxed{F = 149.76 N}$$

0/12

$$f = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$

$$f = \frac{(4\pi \times 10^{-7})(1)(2)(0.002)}{2\pi(0.03)}$$

$$= \frac{16\pi \times 10^{-10}}{0.06\pi}$$

$$f = 2.667 \times 10^{-8} N$$